

SOFTWARE REQUIREMENTS SPECIFICATION

JalMitra

Online Water Supply Management System



Version 1.0

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Under the of Guidance

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Document Information

Item	Details
Project Name	JalMitra – Online Water Supply Management System
Document	Software Requirements Specification
Version	1.0
Platform	Responsive Web Application
Primary Users	Customer, Supplier, Admin
Development Approach	Mobile-first frontend with planned backend/database integration

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Appendix A. Diagrams

1. Introduction

JalMitra is a web-based water supply platform where customers can find nearby water suppliers and order water online.

The main purpose of this project is to make the water ordering process simple and organized. Customers can check supplier details, select water type and quantity, provide their delivery address and place an order.

Suppliers can manage their profile, water details and customer orders. An admin can manage users, suppliers and orders.

2. Problem Statement

Currently, customers usually find water suppliers through phone calls or personal contacts. They may not know the supplier's price, water type, availability or service area.

There is also no simple system to manage customer orders and supplier information in one place.

JalMitra solves this problem by providing one platform for customers and water suppliers.

3. Objectives

The main objectives of JalMitra are:

- To make water ordering easy.
- To connect customers with water suppliers.
- To show supplier and water details clearly.
- To allow customers to place orders online.
- To help suppliers manage their orders.
- To provide an admin panel for managing the system.
- To make the website mobile-friendly.

4. Scope of the Project

JalMitra will mainly focus on local water supply and delivery.

Customer

- Create an account.
- Login to the website.
- Search for suppliers.
- Check supplier details.
- Select water type and quantity.
- Enter delivery address.
- Place an order.
- Select payment method.
- Check order status.
- View previous orders.

Supplier

- Create a supplier account.
- Manage profile.
- Add water details.
- Set price and availability.
- View customer orders.
- Accept or reject orders.
- Update delivery status.

Admin

- Manage users.
- Manage suppliers.
- Verify suppliers.
- Manage orders.
- View basic reports.

5. Users of the System

User	Description
Customer	The customer searches for water suppliers and places orders.
Supplier	The supplier provides water and manages customer orders.
Admin	The admin manages the complete system.

6. Main Modules

A. Customer Module

- Registration
- Login
- Profile
- Supplier Search
- Supplier Details
- Water Selection
- Order Placement
- Payment Selection
- Order Tracking
- Order History
- Reviews & Ratings

B. Supplier Module

- Supplier Registration
- Login
- Supplier Profile
- Water Management
- Price Management
- Availability
- Order Management
- Order Status
- Delivery Management

C. Admin Module

- Admin Login
- Dashboard
- User Management
- Supplier Management
- Supplier Verification
- Order Management

7. Basic Order Flow

JalMitra - Basic Order Flow

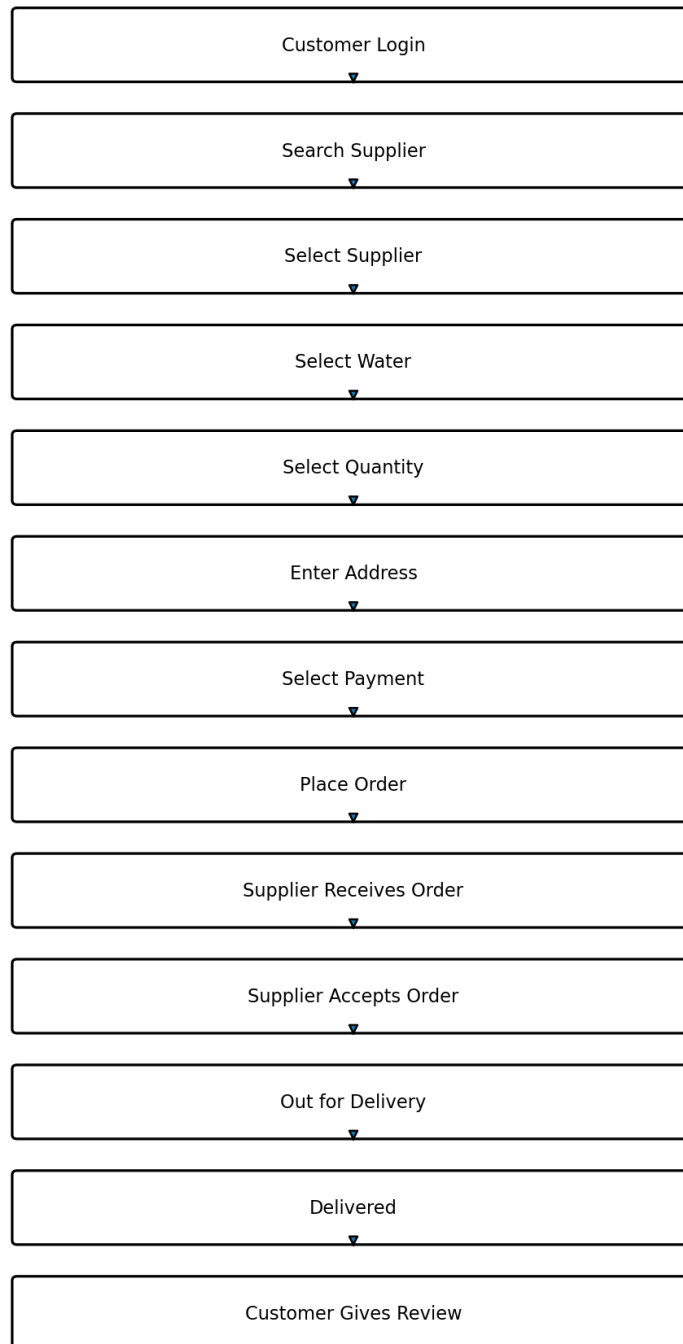


Figure 1: JalMitra Basic Order Flow

8. Technology Used

Part	Technology
Frontend	React.js
Styling	CSS
Backend	Node.js + Express.js
Database	MongoDB
API	REST API
Authentication	JWT
Version Control	Git/GitHub

9. Basic Database

Collection / Table	Main Fields
Users	User ID, Name, Phone, Address,
Suppliers	Supplier ID, Email, Business Name, Phone, Password, Location, Supply Area, Water Type, Capacity
Water Products	Product ID, Supplier ID, Water Type, Quantity, Price, Stock
Orders	Order ID, Customer ID, Supplier ID, Product, Quantity, Address, Payment Method, Amount, Order Status
Reviews	Customer ID, Supplier ID, Rating, Comment

10. System Requirements

Hardware

- Mobile/Computer/Laptop
- Minimum 4 GB RAM
- Internet connection

Software

- Windows/Linux/macOS
- VS Code
- Node.js

- MongoDB
- Web Browser
- Git & GitHub

11. Assumptions

- Customer and supplier will have internet access.
- Suppliers will provide correct information.
- Customer will provide a valid delivery address.
- The system will initially focus on local water delivery.
- Payment options may be expanded in future versions.

12. Limitations

For the first version:

- Live GPS delivery tracking will not be included.
- Advanced payment integration may be added later.
- Delivery service will depend on supplier availability.
- The system will initially focus on a limited service area.

13. Future Scope

In future, JalMitra can be expanded with:

- Current scope :- Live delivery tracking ,Online payment ,Push notifications
- Google Maps integration
- Coupons and offers
- Advanced supplier reports
- Customer support/chat

14. Conclusion

JalMitra provides a simple platform for customers and water suppliers.

It makes the process of finding a supplier, ordering water and managing deliveries more organized.

The system is designed with a mobile-first and responsive approach, so it can be further developed into a mobile application in the future.

Appendix A – System Diagrams

A.1 Use Case Diagram

JalMitra - Use Case Diagram

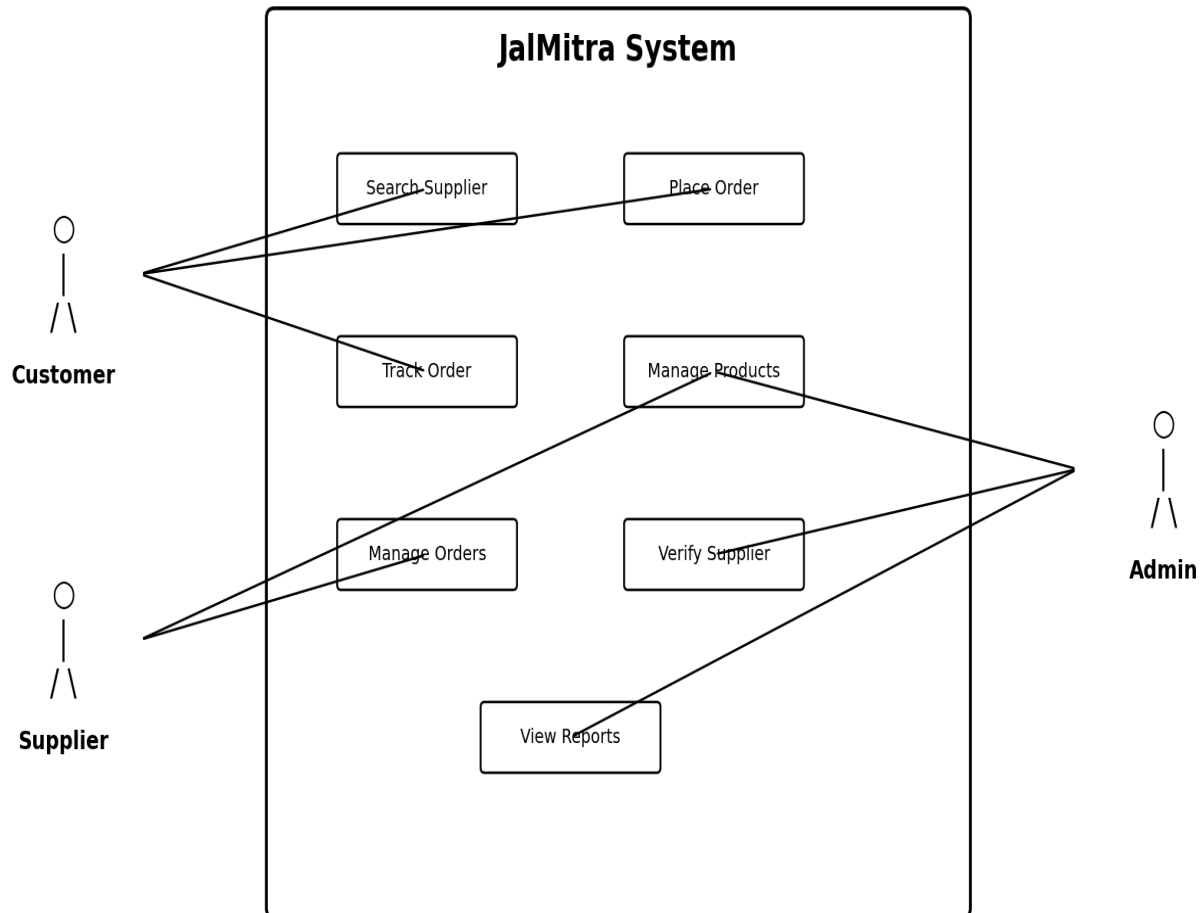


Figure 2: JalMitra Use Case Diagram

A.2 DFD Level 0 / Context Overview

JalMitra - DFD Level 0 / Context Overview

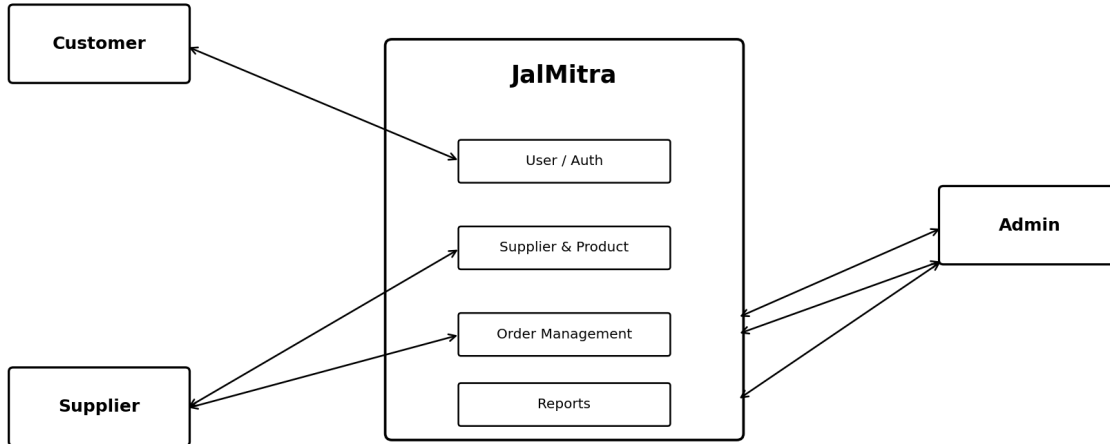


Figure 3: JalMitra DFD Level 0

A.3 ER / Data Relationship Diagram

JalMitra - Basic ER / Data Relationship Diagram

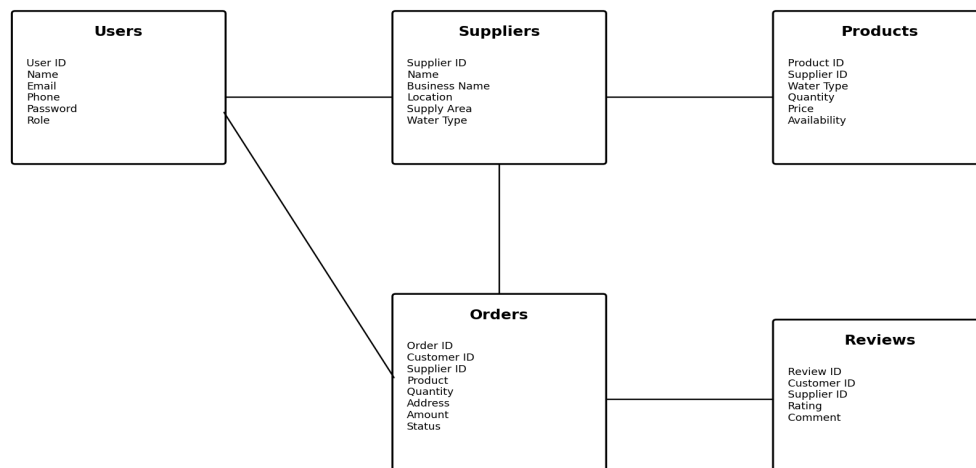


Figure 4: JalMitra Basic ER Diagram

